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Multi-Sensor Fusion and Deep Learning Approaches for Semantic Segmentation of Glacial Lakes: A Comparative Study for Coastal Hydrology Applications

Lingling Xue, Asad Khan, Muhammad Haseeb, Mourad Aqnouy, Dawood Ahmad, Refka Ghodhbani, Dmitry E. Kucher, Olga D. Kucher

***Abstract*—Monitoring glacial lakes is critical for assessing climate change impacts and mitigating glacial lake outburst flood risks. This study evaluates three deep learning architectures, U-Net, Simple Convolutional Neural Network (CNN), and Atrous Spatial Pyramid Pooling SegNet (ASPP SegNet), for binary semantic segmentation of glacial lakes using multi-sensor optical satellite imagery (Sentinel-2). Incorporating data augmentation and custom evaluation metrics (IoU, F1-score, validation loss), the results show that Simple CNN achieves the highest IoU (0.9155) and F1-score (0.9557). At the same time, ASPP SegNet demonstrates superior generalization with the lowest validation loss (0.03337). U-Net also delivers a reliable performance, albeit slightly lower. Visual and quantitative assessments highlight the advantage of multi-scale, context-aware architectures in delineating fragmented lake boundaries. This comparative study provides practical guidance for deep learning model selection in remote sensing-based glacial and coastal hydrology monitoring. Future work will explore temporal modeling, multi-class segmentation, and the integration of optical, radar, and elevation data for improved resilience.**

***Index Terms*—Deep Learning, Sentinel-2 Imagery, Semantic Segmentation, U-Net, Simple CNN, ASPP SegNet, Remote Sensing, Glacial Lake, and Coastal Monitoring**

This work was supported by the National Key Research and Development Program of China (No. 2021YFC3001904 and 2021YFC3090403) and Special Funds for Basic Research Business Fees of China Academy of Safety Science and Technology (No.2023JBKY04). Nanjing Institute of Technology Scientific Research Start Fund (YKJ202118), Natural Science Foundation of Jiangsu Province (BK20201468, 24KJB170010). The authors extend their appreciation to Northern Border University, Saudi Arabia, for supporting this work through project number (NBU-CRP-2025-2461) (Corresponding author: Asad Khan)

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I. INTRODUCTION

Glacial lakes, which are progressively acknowledged as important markers of cryospheric and hydrological change, have been widely formed and expanded as a result of the rising speed of glacier retreat under a warmer climate [1], [2]. These lakes indicate continued glacier mass loss and pose significant risks to downstream communities and infrastructure [3]. For catastrophe risk assessment, water resource management, and long-term climate monitoring in both highland and lowland areas, the timely and precise mapping of glacier lakes is therefore vital [4], [5].

The urgency of monitoring glacial lakes is underscored by the rising frequency and severity of glacial lake outburst floods (GLOFs). According to recent studies, the number of glacial lakes worldwide has increased by approximately 53% since 1990, with total lake area growing by over 51% [6]. GLOFs can release millions of cubic meters of water in a short time, leading to catastrophic downstream flooding, infrastructure damage, and loss of life. For instance, the 2013 Kedarnath disaster in India, linked to glacial lake overflow, resulted in more than 5,000 fatalities [7], [8], [9], [10], [11]. These statistics highlight the critical need for timely, accurate glacial lake monitoring to support early warning systems, hazard assessment, and climate resilience planning[12].

Remote sensing has been a crucial technique for glacial lake discovery and monitoring in recent years, as it offers repeated, large-scale, reasonably affordable observations [13], [14]. However, conventional image classification methods are seriously challenged by the complexity of glacial lake ecosystems, which are defined by changing sunlight, seasonal snow cover, debris-laden ice, and cloud pollution. High altitude and difficult terrain aggravate these restrictions even more as manual demarcation or simply spectral-based methods sometimes yield inconsistent or inaccurate results [15], [16], [17].

While multi-sensor data fusion, involving optical, SAR, and elevation datasets, holds great promise for enhancing hydrological feature delineation, the present study specifically utilizes multi-band optical imagery from the Sentinel-2 satellite. The visible and near-infrared bands (Bands 8, 4, and 3) are employed to maximize water body contrast, and future work will explore the integration of additional modalities such as SAR (e.g., Sentinel-1) and Digital Elevation Models (DEMs) to improve segmentation robustness under variable environmental conditions further [18]. By aggregating complementary geographical, spectral, and temporal information, multi-sensor data fusion improves the depiction of glacial lakes, lowering the ambiguities in single-source datasets. Still, using this abundance of data calls for strong, scalable analytical methods able to understand spatial

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hierarchies and record contextual relationships [19], [20].

Particularly convolutional neural networks (CNNs) intended for semantic segmentation, deep learning models have demonstrated amazing performance in Earth observation activities [21], [22]. Because of its encoder-decoder architecture and capacity for dense pixel-wise categorization, architectures like U-Net and SegNet have grown somewhat well-known[23]. The Atrous Spatial Pyramid Pooling (ASPP) module and other enhancements improve the receptive field of the network, therefore allowing the collection of multi-scale elements essential for precise boundary delineation in challenging terrain [24], [25]. Despite their shown efficiency, the literature lacks a comparative analysis of such designs for glacial lake segmentation, especially concerning real-world multi-sensor satellite data [4], [26].

Tackling these issues requires more advanced analytical tools to catch intricate spatial patterns and reduce human involvement. Recent developments in satellite remote sensing and the rise of deep learning techniques, especially CNNs, provide encouraging paths for automating glacier lake recognition and segmentation with greater accuracy and consistency. Using multi-sensor satellite data and strong machine learning models helps to overcome the constraints of conventional mapping techniques and more closely track these sensitive cryosphere processes related to climate change [27], [28].

In this work, we evaluate three deep learning architectures U-Net, a baseline Simple CNN, and ASPP-enhanced SegNet for the semantic segmentation of glacier lakes using high-resolution satellite images [29]. Model training and assessment based on the publicly accessible Glacial Lake Dataset consists of multi-sensor satellite images and matching binary segmentation masks. To benchmark the strengths and constraints of every method, our study uses conventional performance criteria, including F1-score, Intersection over Union (IoU), and visual quality evaluation[30].

Unlike previous studies that independently applied U-Net or ASPP-based models for glacial lake or snow segmentation tasks, this work uniquely benchmarks three distinct CNN architectures U-Net, Simple CNN, and ASPP SegNet, under a standardized experimental pipeline using a carefully curated multi-sensor glacial lake dataset [31], [32]. This study advances the practical understanding of model behavior in high-altitude hydrological applications by conducting a systematic cross-model comparison based on custom evaluation metrics and qualitative analyses across heterogeneous Himalayan landscapes. This comparative approach provides actionable insights for selecting deep learning models suited to remote sensing–based cryospheric monitoring. This addresses a critical gap in current literature where head-to-head performance evaluations remain limited [33].

Beyond model benchmarking, this work advances the expanding area of automated geospatial intelligence by investigating how contemporary deep learning methods can enable the exact extraction of inland water bodies in dynamic cryospheric settings[34]. With consequences for long-term environmental and coastal monitoring, climate resilience planning, and the development of data-driven decision support systems in Earth observation, the outcomes shed light on the design and choice of deep learning models for remote sensing applications in coastal areas, glacial and high-altitude hydrology[35].

Driven by developments in both remote sensing platforms and image processing techniques, the search for glacial lakes has changed dramatically throughout the past two decades [36], [37]. Applied to satellite images from Landsat, MODIS, and Sentinel missions, glacial lake mapping historically depended on threshold-based spectral indices such as the Normalized Difference Water Index (NDWI), modified NDWI (MNDWI), and automated water extraction index (AWEI) [38], [39], [40], [41]. Although these methods are often useful in places with shadows, snow cover, or murky meltwater, they are prone to misinterpretation. Their reliance on single-sensor input and inflexible rulesets also restricts their flexibility in many environmental circumstances [42].

Many studies have looked at machine learning and deep learning models for water body recognition to go above these constraints. Firstly, Feyisa et al. [43] developed a modified water index algorithm utilizing Landsat images to differentiate water from built-up regions and vegetation. More recently, Shugar et al. [6] underlined the critical necessity of scalable and accurate detection frameworks using a worldwide-scale inventory of glacier lakes and their development, utilizing multi-temporal Landsat data and automated classification algorithms[44].

The rise of CNNs has drastically changed the terrain of remote sensing image processing[45]. First presented U-Net, a completely neural architecture designed for biological image segmentation, Ronneberger et al. [46] have since become somewhat popular in geographical fields. Another study effectively modified U-Net for Himalayan glacier lake detection utilizing Sentinel-2 data, therefore proving better delineation accuracy over conventional techniques in the context of glacial lake mapping [47].

Another encoder-decoder architecture developed by Badrinarayanan et al. [48] has also been used in remote sensing applications. It is usually improved with modules like ASPP to collect multi-scale contextual data better. For water body extraction from high-resolution data, Chen et al. [25] assessed ASPP-enhanced SegNet and DeepLabV3+ architectures, which exhibit appreciable edge preservation and categorization of minor lake features [49], [50].

Apart from model creation, the combination of multi-sensor data has been progressively included into environmental monitoring processes [51]Combining multispectral optical data with auxiliary sensors like SAR, LiDAR, and digital elevation models (DEMs) helps lower uncertainty brought on by cloud cover, changing sun angles, or spectral confusion. Nguyen et al. demonstrate the value of multi-source input in water-related applications. [52] suggested a hybrid deep learning method that combined Sentinel-1 SAR with Sentinel-2 optical data for land-cover mapping in flood-prone regions. Elevation and temperature information can also enhance optical data in glacier areas, strengthening model resilience for differentiating shadows from ice-covered zones and lakes.

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Deep learning has been used recently to investigate cryospheric properties outside of lake detection. Whereas Xie et al. [53] utilized a hybrid CNN–LSTM network for time-series analysis of glacier meltwater characteristics, Yin et al. [54] used a YOLOv5-seg network framework for snow cover segmentation on the High Mountain Asia [55]. These investigations show the increasing relevance of semantic segmentation models in complicated high-altitude and cold-region settings, where glacier lakes are fundamental hydrological agents. Nevertheless, despite these developments, thorough benchmarking studies evaluating the performance of several CNN architectures under consistent settings for glacial lake segmentation are still lacking, which this work attempts to solve [56], [57].

Regarding datasets, the most recent Glacial Lake Dataset published on Kaggle by Shrestha [58] offers a benchmark resource with well-chosen segmentation masks for supervised deep learning and multi-sensor satellite images [59]. Comparative research across model architectures and repeatable studies have been made possible by this dataset. Besides lake segmentation, many general geoinformatics studies have underlined multi-sensor data fusion methods. Establishing fundamental models for combining multispectral and panchromatic data, Pohl and Van Genderen [60]. More recent work by Zhu et al. [61] has used data fusion with deep neural networks for better land cover categorization across varied terrain. Despite these advances, relative assessments of CNN-based segmentation models in the particular setting of glacial lake identification remain few [57], [62]This work fills this need by assessing U-Net, a simple CNN, and ASPP-enhanced SegNet on a carefully chosen multi-sensor dataset, Focusing on segmentation accuracy, model stability, and practical application in cryospheric conditions.

Despite the significant progress made in remote sensing of glacial lakes through both classical and deep learning approaches, a consistent comparative evaluation of modern segmentation architectures using standardized, high-resolution, multi-sensor datasets remains largely underexplored. Most existing studies focus on single-model implementations or rely on region-specific case studies, limiting the generalizability and reproducibility of their findings. Furthermore, although promising, integrating multi-sensor spectral information with deep neural networks requires further investigation to fully exploit spatial, spectral, and contextual cues in complex glacial environments. Our study addresses this gap by systematically benchmarking three representative deep learning models, U-Net, a Simple CNN, and ASPP SegNet on a curated glacial lake dataset. The results aim to guide future model selection and application design for large-scale hydrological and cryospheric mapping in remote sensing workflows.

II. Material and Methods

The dataset properties, preprocessing techniques, and deep learning architectures used for the semantic segmentation of glacier lakes from satellite images are presented in this part. The proposed technique combines high-resolution multi-sensor images with three CNN models: U-Net, a bespoke Simple CNN, and ASPP SegNet to evaluate their efficacy in defining glacial lake borders under different spectral and spatial conditions. The publicly accessible Glacial Lake Dataset has been used for the tests; it comprises annotated satellite pictures spanning several geographical areas. Using a uniform data pipeline, every model has been trained and verified under evaluation using conventional semantic segmentation criteria. The dataset construction, preprocessing techniques, network settings, and training framework applied in this comparison study are covered in the next subsections.

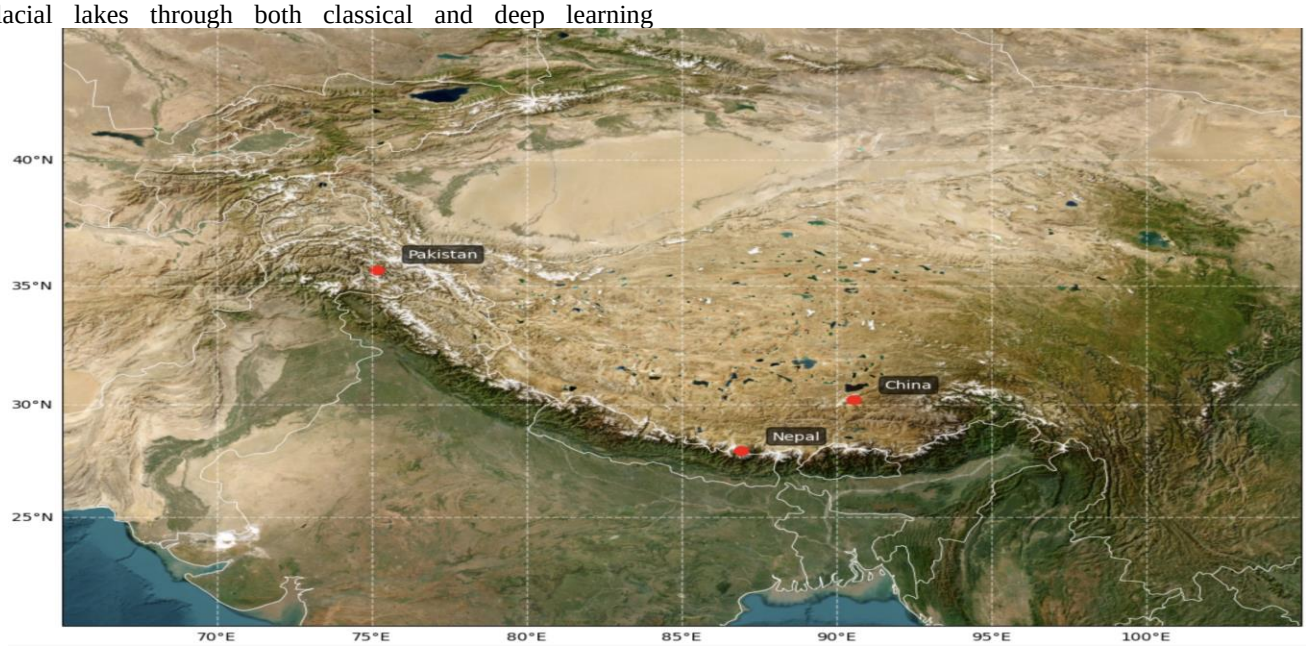

Fig 1. Study area location map showing the selected glacial lake regions across high-altitude Himalayan environments. The figure illustrates the geographic distribution of sampling sites, chosen to represent diverse environmental conditions, including varying altitudes, land cover types, and lake morphologies, ensuring the robustness of segmentation model evaluation.

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A. Study Area and Dataset

This work focuses on glacial lake ecosystems inside the Himalayan region, covering high-altitude areas of Nepal, China, and Pakistan (Fig 1). Often referred to as the "Third Pole," the Himalayas are among the most important cryospheric areas in the world as their vast glacier sustains downstream hydrology and ecosystems and feed important transboundary river systems [63]. Emphasizing their larger relevance in regional water resource dynamics, these alpine zones reflect both the headwaters of coastal-draining basins and a supply of freshwater for highly populated areas downstream [64], [65]. Driven by rapid glacier melt and periodic hydrometeorological variability, the development and spread of glacial lakes in these areas has become a major sign of climate change and a possible cause of glacial lake outburst floods. Strong seasonal and regional fluctuation in size, turbidity, and spectral properties among these dynamic inland water bodies provides a challenging but pertinent instance for improving remote sensing–based water segmentation techniques [63], [65].

The Sentinel-2 satellite constellation, as shown in Fig 2, a Copernicus Earth observation mission component, provides the images utilized in this work. Comprising carefully chosen picture patches with a spatial resolution of 400 × 400 pixels, the dataset makes use of false-colour composites grounded on Bands 8 (near-infrared), 4 (red), and 3 (green) [66]. Using spectral contrast throughout visible and near-infrared wavelengths, this multi-spectral arrangement improves water body delineation and helps to identify glacial lakes from surrounding snow, ice, and bare ground. Using multi-sensor optical data helps to assist the more general goal of providing feature-level fusion for high-precision mapping across difficult mountainous terrain [64].

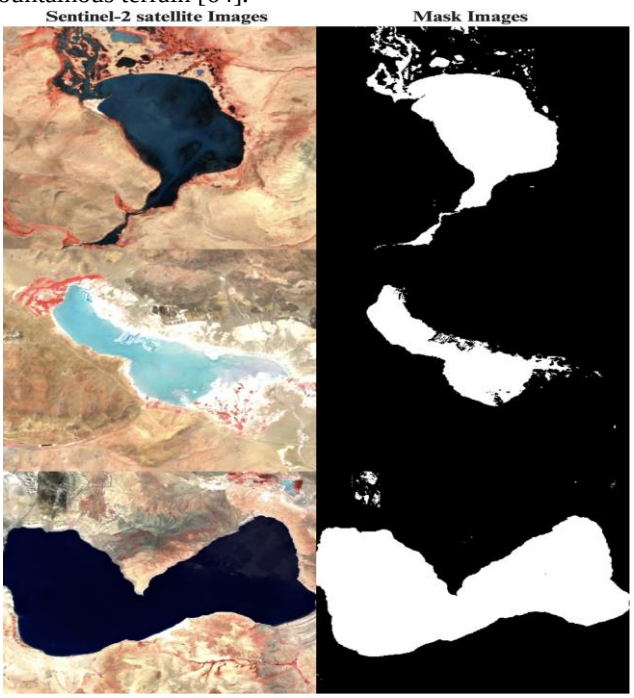

Fig 2. Examples of input Sentinel-2 multi-band composite imagery alongside manually curated ground truth segmentation masks used for model training and evaluation. The figure highlights the spectral variability of glacial lakes across different sites and the challenges posed by shadowing, snow contamination, and small, fragmented lake features that require careful delineation.

Every image tile is matched with a binary lake mask that offers pixel-wise glacial lake boundary annotations, enabling the dataset to be fit for supervised semantic segmentation projects. Evaluating the effectiveness and generalizability of deep learning models in challenging hydrological contexts also finds a strong basis in the variety of landforms, atmospheric conditions, and seasonal surface states shown in the dataset. Although the study area is inland mountainous, the hydrological relevance of glacial lakes to downstream coastal systems and their significance in regional water transfer paths firmly place this research inside the scope of multi-sensor remote sensing for coastal and watershed monitoring.

B. U-Net Architecture

The U-Net model employed in this work uses a fully convolutional neural network, especially intended for exact semantic segmentation of glacier lakes and coastal borders using Sentinel-2 satellite images [37]. Applied with TensorFlow and Keras, the model uses symmetric skip connections in an encoder-decoder architecture to efficiently preserve spatial information throughout the network.

Four successive 3x3 convolutional layers with ReLU activation functions make up the encoder route, which runs twice two max-pooling operations. From 64 to 512 across encoding layers, the number of feature channels is gradually raised to enable hierarchical extraction of spectral and spatial data, including those related to shadowed lake borders and difficult topography reliefs. Considered the deepest feature representation, the bottleneck layer consists of two 3x3 convolutions with 1024 filters. Following 2x2 upsampling layers and concatenation with matching encoder feature maps via skip connections, the decoder path replicates the encoder. These skip connections reduce pooling-induced feature loss and maintain fine-grained boundary information. At every decoding level, later 3x3 convolutional layers hone the upsampled features.

Using a 1x1 convolutional layer with sigmoid activation, the last segmentation map generates a binary output mask denoting glacial lake existence. A binary cross-entropy loss function tuned using the Adam optimizer trains the model. TensorFlow's subclassed Metric API is used to create custom evaluation metrics like the mean IoU and F1-score, offering strong performance evaluations throughout the 25-epoch training phase. 400x400 pixel tiles generated from the Sentinel-2 dataset (Bands 8, 4, 3), normalized to [0, 1], make up input data. Random horizontal and vertical flips are part of the training process to improve generalization and stop overfitting. Every image–mask pair also undergoes a special operation to expand the training set artificially.

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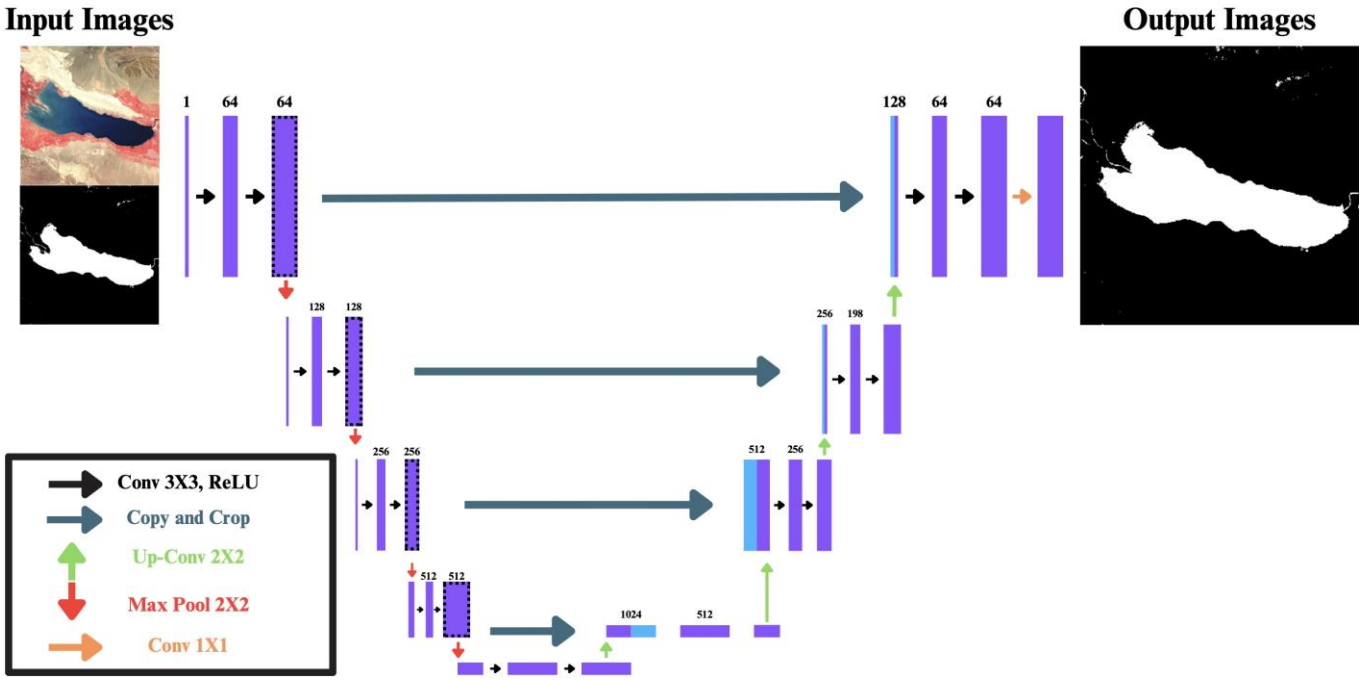

Fig 3. U-Net architecture used for glacial lake and coastal segmentation.

C. *Simple CNN Architecture*

The shallow CNN [67] has been built and trained for glacial lake extraction to offer a lightweight baseline for semantic segmentation. Acting as a benchmark for trade-offs between model complexity and segmentation efficiency, this Simple CNN model provides a computationally efficient substitute for more complicated designs such as U-Net and ASPP SegNet.

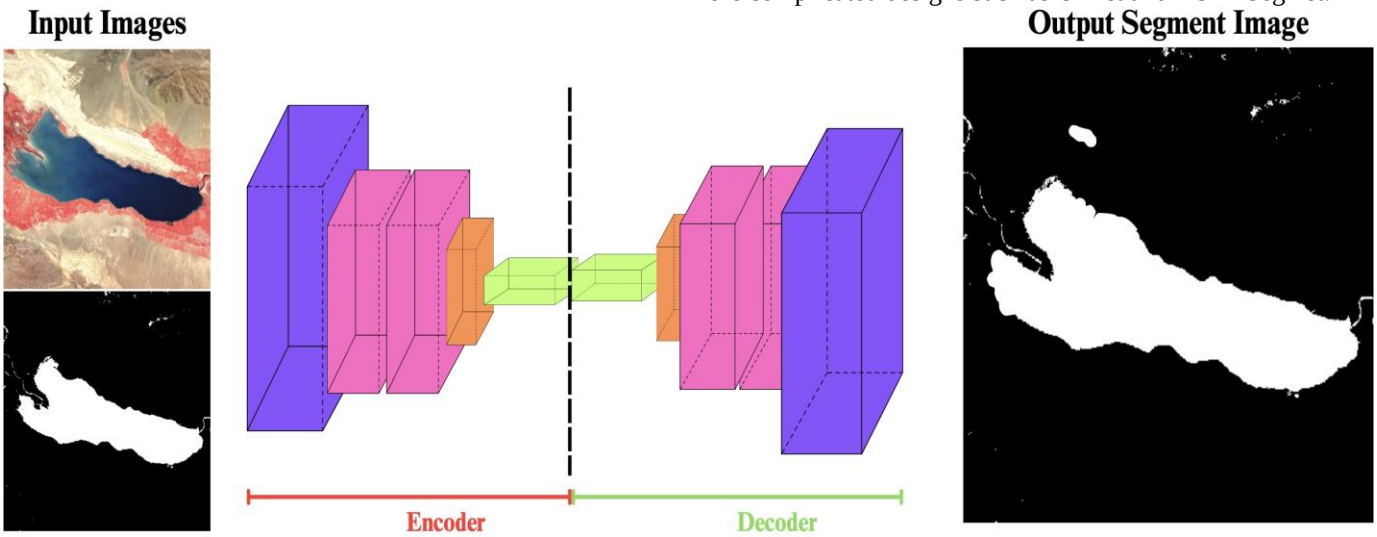

Fig 4. Schematic of the Simple CNN architecture designed for glacial lake segmentation. The model consists of a sequential encoder–decoder structure with convolutional and pooling layers in the encoder and upsampling layers in the decoder.

Comprising a sequential stack of convolutional and pooling layers meant to collect and rebuild spatial data with little parameter overhead, the model starts with a 3x3 convolutional layer with 32 filters; the encoder downsamples the spatial dimensions using a ReLU activation function and a 2x2 max pooling layer. To maintain the same convolution and pooling structure, two more convolutional blocks with 64 and 128 filters, respectively, carry on the feature-extracting process. Using form, texture, and spectral contrast from the input Sentinel-2 images, this configuration helps the network to learn basic spatial patterns linked with glacier lakes.

The decoder consists of two upsampling layers (2x2), each followed by a convolutional layer with decreasing filter widths (64 and 32), therefore enabling slow recovery of spatial information and reconstruction of the original resolution of the segmentation mask. A 1x1 convolution with a sigmoid activation generates a binary segmentation mask matching water pixel presence from the output layer. All layers apply the

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"same" padding for dimensional alignment during the forward pass. Accepting 400x400x3 input tiles, the model generates a 400x400x1 probability map thresholded at 0.5 for binary classification.

Like the previous models, the Simple CNN is trained with binary cross-entropy loss and then refined with an Adam optimizer. TensorFlow used custom measures like F1-score and IoU to assess the model's capacity to separate lake areas from the surrounding landscape precisely. To increase resilience to orientation and spatial variation, the training process additionally uses synthetic duplication and random flipping, thereby augmenting data. A Simple CNN has been deliberately selected as a baseline model to establish a reference point for assessing the added value of architectural complexity in segmentation tasks. Shallow networks, characterized by fewer layers and parameters, offer faster training times, lower computational demands, and are more feasible for deployment in real-time or resource-limited environments. By comparing the Simple CNN with more sophisticated architectures like U-Net and ASPP SegNet, this study aims to comprehensively understand the trade-offs between model simplicity, accuracy, and generalization, thereby supporting more informed model selection decisions for operational glacial hazard monitoring.

D. Atrous Spatial Pyramid Pooling SegNet Architecture

We developed a modified SegNet architecture enhanced with an ASPP module to improve the model's capacity for capturing multiscale spatial patterns across diverse glacial lake geometries [24], [39]. Particularly suitable for segmenting lakes with erratic borders and varied sizes in remote sensing images, the incorporation of ASPP helps the model integrate contextual information efficiently at several receptive fields.

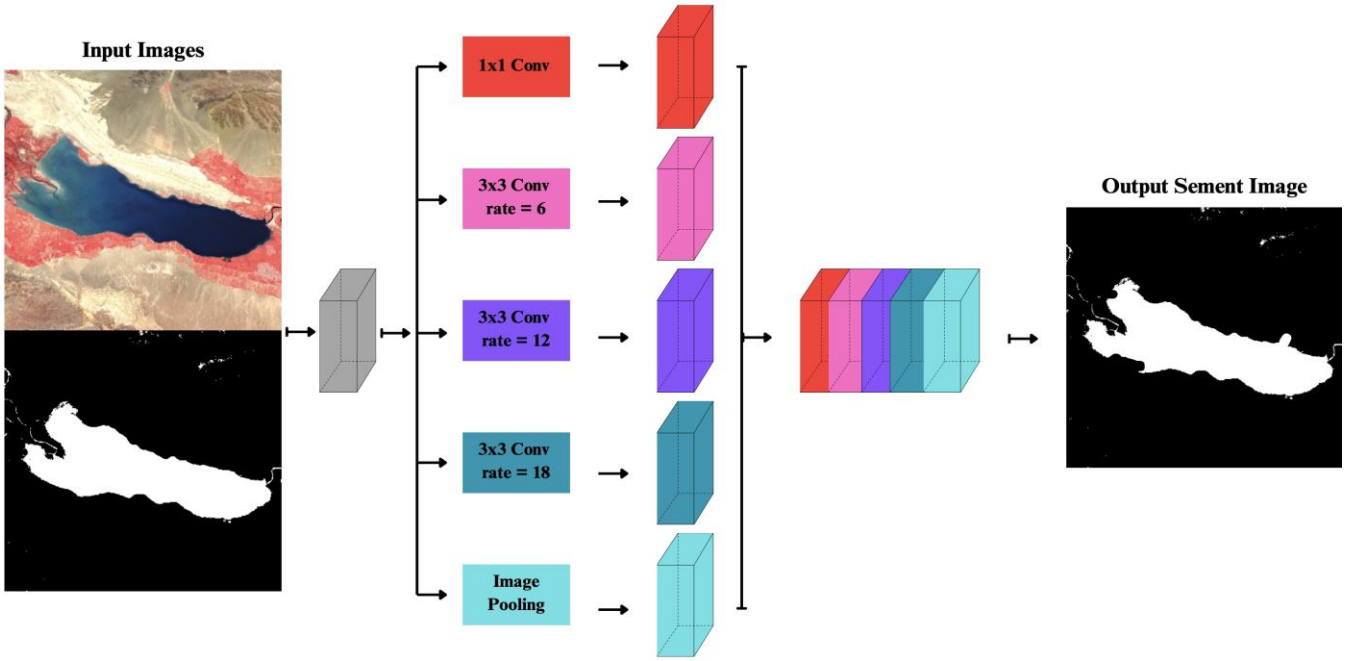

Fig 5. Architectural schematic of the Atrous Spatial Pyramid Pooling (ASPP) module integrated within the SegNet framework

Starting with a three-convolutional block conventional SegNet-style encoder, the architecture proceeds. Two successive 3x3 convolutional layers with ReLU activation make up each block; then, a 2x2 max-pooling operation downsamples. From 64 to 256, the number of filters rises gradually, allowing hierarchical feature extraction and lowering the spatial resolution of the input feature maps. The model presents the ASPP module after the encoder using a 1x1 convolution branch and parallel dilated convolutions with varying dilation rates (6, 12, 18). Global average pooling provides an extra image-level context branch; thereafter, a 1x1 convolution and bilinear up-sampling bring the original resolution back. A 1x1 convolution polishes the concatenated output of these branches even further, generating a feature map loaded with both local texture and global context. This approach lets the model find bigger, more continuous water bodies with great precision and tiny, scattered glacier lakes.

Two consecutive up-sampling steps let the decoder reflect the encoder. Every up-sampling operation follows a convolutional layer that gradually rebuilds the segmentation mask's spatial resolution. A 1x1 convolution with a sigmoid activation generates a binary segmentation output at the last layer. Sentinel-2 picture patches scaled to 400x400 pixels and normalized to [0, 1] are the input for this model, as with the others. Custom F1-score and IoU measures have been used to evaluate the network after it is trained with binary cross-entropy loss using an Adam optimizer. Data augmentation methods like flipping and picture duplication have been used during training to guarantee resilience across several spectral and topographic circumstances.

E. Comparative Framework for Model Evaluation

A single experimental pipeline has been constructed to methodically evaluate the three semantic segmentation models—U-Net, Simple CNN, and ASPP SegNet—ensuring uniform data preparation, augmentation, training, and evaluation methods. Using false-colour composites (Bands 8,

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4, 3) with spatial dimensions of 400×400 pixels and their matching binary masks representing glacial lake extents, all models have been trained on the same Sentinel-2 picture dataset.

With a batch size of 8, every model has been trained separately using the Adam optimizer and binary cross-entropy loss function 25 times. Two bespoke measures based on TensorFlow's Metric class—the F1-score and the IoU have been used to assess segmentation quality. Calculated per epoch for both training and validation sets, these measures allowed a constant comparison of model convergence behaviour and generalization capacity. Every model received evenly applied random horizontal and vertical flipping as data augmentation methods to boost the variety of training samples and lower overfitting. Further extending the training set and simulating spectral-spatial variability seen in high-altitude glacier conditions has been accomplished using a new data copying feature[68], [69]. The three designs have somewhat different receptive field capabilities and complexity:

- U-Net uses a deep encoder-decoder structure with skip links to merge fine-grained spatial data with abstract semantic information, which proved especially successful in detecting uneven and fractured lake borders.
- Especially in cases with well-contrasted water bodies, Simple CNN provides a computationally efficient, minimalistic design with fewer layers and parameters, yet still achieves competitive performance.
- With an Atrous Spatial Pyramid Pooling module, ASPP SegNet expands the conventional SegNet architecture and lets the model incorporate multiscale context via dilated convolutions. Large or geographically varied glacial lakes with many receptive fields improve feature recognition benefits, especially from this layout.

This work enables a fair and repeatable comparison of the three designs by keeping constant experimental conditions. Apart from showing trade-offs between model complexity and accuracy, the comparison analysis guides model choice for practical purposes in glacial hazard assessment and water resource monitoring.

F. Evaluation Metrics

To quantitatively assess the segmentation performance of the U-Net, Simple CNN, and ASPP-SegNet architectures, two key evaluation metrics have been utilized: the F1-score and the IoU [70]. These metrics are widely employed in semantic segmentation tasks to evaluate the overlap between predicted masks and ground truth labels, especially in binary classification problems such as water body delineation. The IoU, also known as the Jaccard Index, measures the ratio of the area of overlap between the predicted mask and the ground truth mask to the area of their union. It is defined as equation 1 [71]:

$$IoU = \frac{True\ Positives}{True\ Positive + False\ Positve + False\ Negatives} \quad (1)$$

Where TP is the number of true positive pixels, FP is the number of false positives, and FN is the number of false negatives. IoU provides a robust indication of segmentation accuracy, especially for imbalanced datasets.

The F1-score, also known as the Dice Coefficient, is the harmonic mean of precision and recall, providing a balanced measure of the model's performance concerning both over-segmentation and under-segmentation. It is expressed as equation 2 [72]:

$$F1\ Score = \frac{2\ .TP}{2\ .TP + FP + FN} \quad (2)$$

This metric emphasizes both correct identification and coverage of the target class (coastal areas and glacial lakes). The F1 score is particularly useful in cases of class imbalance, where water bodies may constitute a small portion of the image. During training, all models were optimized using the binary cross-entropy loss, which quantifies the pixel-wise difference between predicted probabilities and the binary ground truth labels. The loss is calculated as equation 3:

$$\mathcal{L}_{BCE} = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log(p_i) + (1 - y_i)\log(1 - p_i)] \quad (3)$$

In the binary cross-entropy loss function, $y_i \in \{0,1\}$ represents the ground truth label for each pixel i, while $p_i \in [0,1]$ denotes the predicted probability of that pixel belonging to the foreground class (i.e., glacial lake). The total number of pixels in the image is denoted by N [73], [74]. This formulation penalizes incorrect predictions by measuring the divergence between predicted probabilities and actual labels at the pixel level. By minimizing this loss, the model is guided to produce probability maps that closely match the binary ground truth masks, ultimately facilitating accurate pixel-wise classification of glacial lakes.

Metrics have been computed on both the training and validation datasets across all epochs to track learning progress and generalization ability. The custom implementations of MeanIoU and MeanF1 classes in TensorFlow ensure per-epoch accumulation and correct averaging over batches. Visual inspection of segmentation outputs, in conjunction with numerical evaluation, supported model interpretability and robustness verification.

III. Results And Discussion

These three deep learning architectures U-Net, Simple CNN, and ASPP SegNet used for glacier lake segmentation from Sentinel-2 imagery are comprehensively evaluated in this part. Using both quantitative measures and visual examination, the models have been assessed to determine their capacity to precisely define glacial lake margins under various geographic and spectral contexts. Over 25 seasons, key performance measures, including F1-score and IoU, have been examined across training and validation datasets. The comparison results offer important new perspectives on each model's strengths and constraints in addressing spectral variation, generalizing over many Himalayan research locations, and capturing fine-scale lake form. The conversation then places these results in perspective concerning coastal zone analysis, climate change assessment, and geospatial applications in hydrological monitoring.

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TABLE I
TRAINING CONFIGURATION AND HYPERPARAMETERS USED FOR U-NET, SIMPLE CNN, AND ASPP SEGNET MODELS.

Parameter	Value
Number of Epochs	25
Batch Size	16
Optimizer	Adam
Learning Rate	0.001
Loss Function	Binary Crossentropy
Input Image Size	256 × 256 pixels
Data Augmentation	Rotation, Flipping, Scaling
Evaluation Metrics	IoU, F1-score, Accuracy
Framework	TensorFlow/Keras

Based on Table I, all three models have been trained under identical conditions to ensure a fair comparison. The Adam optimizer has been selected for its adaptive learning capabilities and a binary cross-entropy loss function suitable for two-class segmentation tasks (glacial lake vs background). The dataset has been resized to 256x256 pixels to balance detail and computation time. Based on preliminary experiments for efficient convergence, a batch size of 16 and 25 training epochs has been chosen. Data augmentation techniques, including random rotations, horizontal and vertical flips, and scaling, have been applied to increase training diversity and improve generalization. The training has been implemented in TensorFlow/Keras, and evaluation has been conducted using custom metrics, IoU and F1-score, alongside pixel-wise accuracy. Inference speed has also been recorded to compare the computational efficiency of the models.

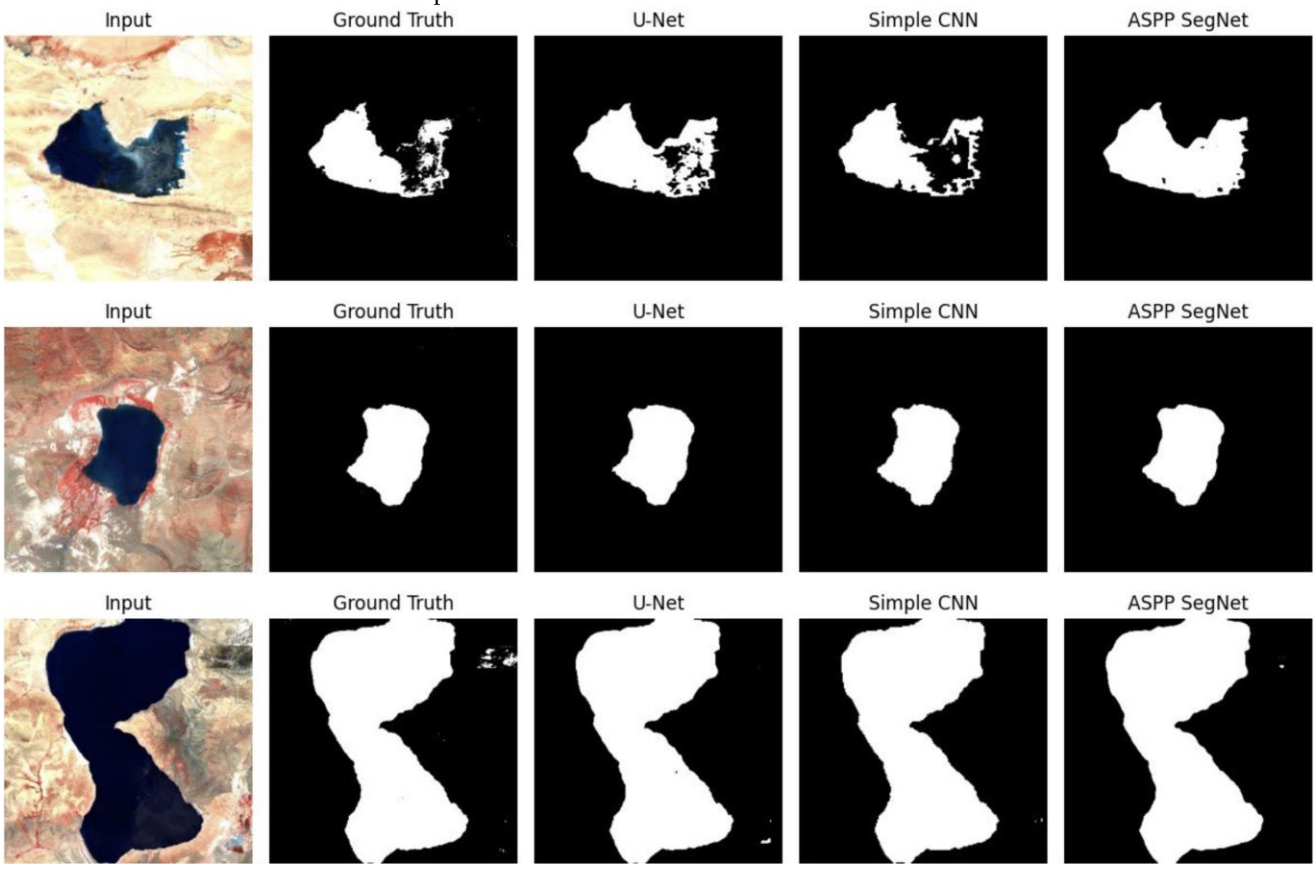

Fig 6. Comparative visualization of glacial lake segmentation results using U-Net, Simple CNN, and ASPP SegNet architectures on Sentinel-2 imagery.

Fig 6 shows qualitative segmentation results generated by U-Net, Simple CNN, and ASPP SegNet across three example Sentinel-2 picture patches spanning various glacial lake habitats. Every row displays the three models' segmentation predictions, the ground truth mask matching the input satellite picture, and important new perspectives on the spatial correctness and generalization capacity of the investigated architectures come from this visual comparison.

All three models in the first row effectively define the lake border; U-Net and ASPP SegNet show better edges and closer conformance to the actual lake extent. Although it detects the general form, the Simple CNN exhibits slight border abnormalities and internal artefacts, probably resulting from its shallow design and poor feature extraction depth. The second sample shows a more complicated glacier environment, including diverse landscape textures and cloud-like elements. Here, using its multi-scale receptive field, ASPP SegNet performs better by keeping a distinct separation between the lake and adjacent snow-covered regions. Though with small over-segmentation close to high-reflectance limits, U-Net also works well. In this scenario, a simple CNN struggles more obviously and misclassifies terrain features as water.

The third row shows a vast, continuous lake body with quite homogeneous spectral characteristics. While Simple CNN

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exhibits some under-segmentation at shoreline extremes, all three models, in this case, accomplish high-quality segmentation; ASPP SegNet once displays the clearest border preservation. Particularly in diverse situations with fine-scale complexity, the qualitative comparison generally shows that ASPP SegNet regularly generates higher accurate segmentation outputs. Because of its skip connections and deep design, the U-Net model works well in most situations; occasionally, it is prone to minor edge distortions. Although computationally economical, a simple CNN demonstrates limited capability in addressing spectral and spatial variability, highlighting the trade-off between model simplicity and segmentation precision.

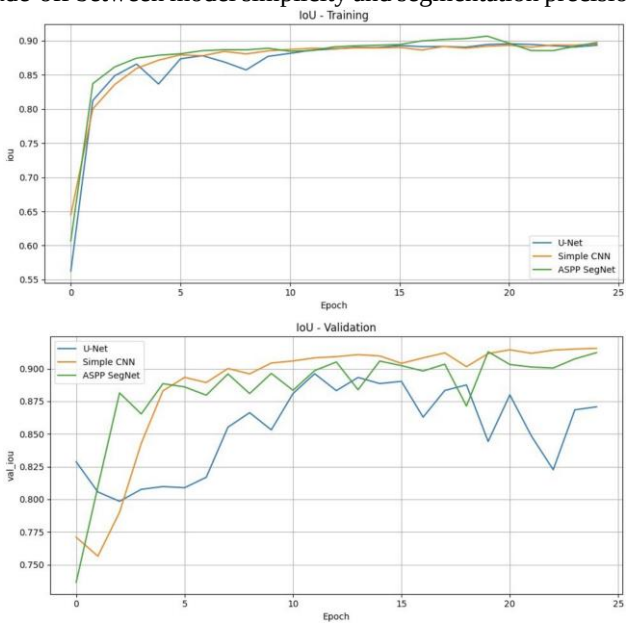

Fig 7. Visual comparison of segmentation outputs from U-Net, Simple CNN, and ASPP SegNet models against ground truth masks.

Fig 7 shows example segmentation results from U-Net, Simple CNN, and ASPP SegNet overlaced with matching ground truth masks to visually assess model performance. These findings shed light on how each design detects Sentinel-2 glacier lake limits. Often producing the most accurate segmentations, U-Net closely matched the ground truth's form and limits. Its encoder-decoder design allowed it to retain spatial resolution and edge features by use of skip connections, therefore reducing false positives and improving lake delineation.

Figs. 6 and 7 observably show key segmentation differences, particularly along glacial lake boundaries and fragmented regions. While no inset enlargements are provided, careful inspection of boundary areas reveals that models with multiscale context capture (e.g., ASPP SegNet) exhibit smoother and more complete lake edge delineations than the shallower Simple CNN. These visual results underscore the significance of boundary-sensitive segmentation for reliable glacial lake monitoring applications.

Particularly in collecting multi-scale lake characteristics, ASPP SegNet also performed well. Though the multi-scale receptive field from the ASPP module suggests over-segmentation in complicated topographic backdrops, its predictions occasionally exhibited over-segmentation in these settings, most likely due to the coarse pooling from its deeper layers. By comparison, Simple CNN showed uneven predictions and suffered from boundary precision. Its shallow design cannot generalize spatial patterns in varied contexts, so it tends to miss smaller lakes and misclassifies topographical features as water bodies. These visual variations match the quantitative measures presented in Table I; U-Net obtained the greatest IoU and F1-score, followed by ASPP SegNet and Simple CNN. A simple CNN provided faster inference. However, this came at a far lower segmentation quality.

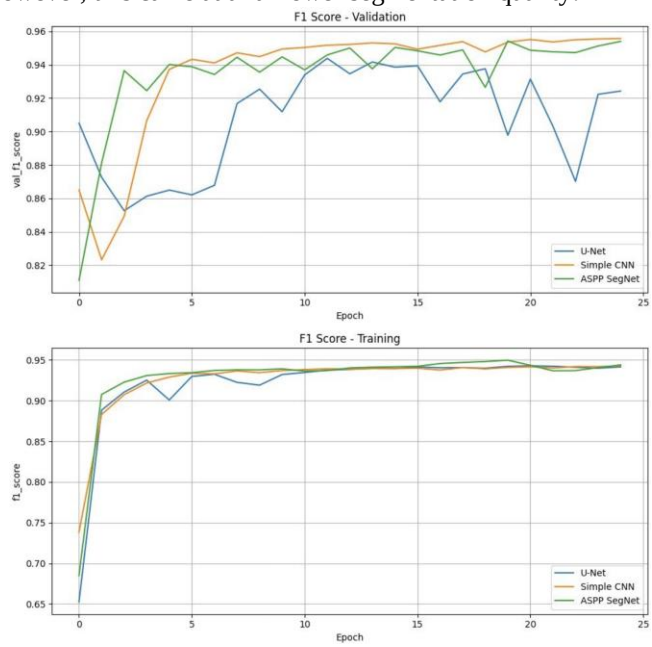

Fig 8. Training and validation of F1-score trends for U-Net, Simple CNN, and ASPP SegNet over 25 epochs.

Intersect over Union (IoU) and F1-score measures on both the training and validation datasets assessed the three segmentation models, U-Net, Simple CNN, and ASPP SegNet. Fig 8 shows the training and validation F1-score curves spanning 25 epochs for every model. Especially in terms of validation F1-score, the data clearly show that ASPP SegNet routinely beats the other designs. Although all models demonstrate convergence, Simple CNN displays variations in validation performance, probably because of its shallow design and limited feature extraction capacity. U-Net runs more stably thanks to its symmetric encoder-decoder architecture and skip connections. However, ASPP SegNet shows better generalization as its ASPP module efficiently records multi-scale contextual information.

Visual confirmation of these results comes from the segmentation outputs shown in Fig 6. With less noise and fewer false positives in challenging settings, ASPP SegNet offers a better definition of glacial lake borders. Simple CNN highlights the limits of simple architecture in high-variability glacier situations by occasionally misclassifying nearby topography as water bodies. Further confirming these results is a quantitative comparison (see Table II). With the greatest IoU and F1-score values, ASPP SegNet shows how well it manages the fundamental difficulties of satellite-based glacial lake

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segmentation, including spectral similarity and fractured lake forms. The results confirm the need to use sophisticated, context-aware structures in remote sensing segmentation activities. Striking a compromise between accuracy and computational economy, ASPP SegNet proves to be the most consistent model in our work.

TABLE II
COMPARISON OF BEST VALIDATION F1-SCORE, IOU, AND LOSS ACROSS U-NET, SIMPLE CNN, AND ASPP SEGNET MODELS.

Model	Best Val F1 Score	Best Val IoU	Best val_loss
UNet	0.9438	0.8961	0.03898
Simple CNN	**0.9557**	**0.9155**	0.0361
ASPP	0.9542	0.9129	**0.03337**

Table II summarizes every model's quantitative performance measures. Simple CNN somewhat beats ASPP SegNet (F1-score: 0.9542; IoU: 0.9955) with the highest validation F1-score, 0.9557. ASPP SegNet obtains the lowest validation loss (0.03337), implying steadier learning and improved generalisation. U-Net displays interesting competitive performance with a validation F1-score of 0.9438 and an IoU of 0.8961. Both measures, nevertheless, laggard the other two models somewhat. Furthermore, its larger validation loss indicates more prediction uncertainty in complicated areas of the input photos (0.03898).

The observed discrepancy between the Simple CNN achieving the highest validation F1-score, yet higher validation loss compared to ASPP SegNet can be attributed to differences in model complexity and generalization behavior. The shallow Simple CNN tends to overfit to more evident and high-contrast water body features, which inflates its validation F1-score on familiar patterns but results in less confident predictions for complex or ambiguous cases, as reflected in its higher validation loss . This suggests that while the Simple CNN may achieve higher pixel-level overlap in ideal scenarios, it lacks robustness against spectral ambiguity, fragmented morphologies, and subtle boundary transitions. In contrast, the ASPP SegNet, equipped with multiscale contextual modules, captures diverse spatial hierarchies and better manages generalization across heterogeneous glacial lake environments. This results in slightly lower peak F1-scores but a more stable and reliable prediction behavior. This highlights the importance of considering both performance metrics and generalization indicators (e.g., validation loss trends) when evaluating model suitability for real-world deployment[75], [76], [77], [78], [79].

Simple CNN leads in raw F1 and IoU measurements; its somewhat greater validation loss (0.0361) compared to ASPP SegNet suggests overfitting to the training data. Integration of atrous convolutions with multi-scale context by ASPP SegNet seems to provide a more balanced and stronger learning process. These results, taken together, highlight the trade-off between metric maximizing and model generalization. Simple CNN scored well on the validation set in terms of F1-score and IoU. Still, ASPP SegNet's better loss minimizing and consistent validation curves (see Fig 8) make it a solid contender for operational deployment in glacial lake segmentation tasks under varied picture circumstances.

The comparison study emphasizes, in general, the advantages and shortcomings of every model architecture. Simple CNN has somewhat better F1-score and IoU values, but its performance is matched by a slightly larger validation loss, indicating perhaps overfitting. Though it shows good baseline performance, U-Net lacks the depth and contextual awareness required for the most precise segmentation. Conversely, ASPP SegNet's lowest validation loss and consistent training behaviour show that it combines great generalization with high accuracy. These results imply that, using Sentinel-2 images, ASPP SegNet is the most appropriate architecture for dependable and scalable glacial lake segmentation. The findings give important new perspectives on model choice for further studies and operational monitoring of glacier lake dynamics in remote sensing applications.

Despite the promising results demonstrated in this study, several limitations should be acknowledged. One notable challenge is the potential for misclassification under cloud cover conditions, snow contamination, and variable illumination, which can introduce spectral ambiguities affecting segmentation accuracy. Additionally, all models may not consistently capture small or highly fragmented glacial lakes, particularly when spectral contrast with the surrounding terrain is low. Future work should further explore integrating cloud-masked datasets, multimodal inputs (e.g., SAR data), and uncertainty quantification techniques to enhance model reliability in complex environmental conditions. Future research might also look at how auxiliary and multimodal datasets could improve glacier lake segmentation results. In consistently overcast or low-illumance settings, Sentinel-1 synthetic aperture radar (SAR) data, which is cloud cover insensitive and can record surface roughness, can augment optical observations. While thermal infrared photos might enable identification of water bodies depending on temperature differences, Digital Elevation Models (DEMs) could help to separate lakes from darkened or snow-covered land by include topographic context. Combining these modalities employing feature-level or decision-level fusion techniques will allow more robust and precise delineation across several environmental situations, increasing deep learning models' relevance in operational cryospheric monitoring and coastal hazard assessment.

IV. CONCLUSION

This work examined the performance of deep learning architectures, U-Net, Simple CNN, and ASPP SegNet, for the semantic segmentation of glacial lakes using Sentinel-2 satellite data. Using pixel-wise classification methods, we sought to meet the increasing need for precise and automated mapping of coastal regions and glacial lakes, essential markers of climate change and possible glacial lake outburst flood dangers.

A strong training and assessment pipeline includes 25-epoch training sessions, data augmentation techniques, and bespoke evaluation metrics catering to the binary segmentation problem. With validation F1-scores over 0.94, the data show that all three models performed very well. Simple CNN fared somewhat better than the others regarding F1-score (0.9557) and IoU

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(0.9955). However, ASPP SegNet showed the highest generalizing capacity, demonstrated by the lowest validation loss (0.03337). Though it is somewhat poor at collecting finer spatial elements and complicated boundaries, U-Net still performs well.

The relative results draw attention to important architectural concessions[80]. Simple CNNs' inclination toward overfitting underscores the necessity of more sophisticated regularization or context-capturing techniques, even if they provide a lightweight design and good peak accuracy. Through ASPP, on the other hand, ASPP SegNet gains from the ability to efficiently handle multi-scale features, a necessity in distant sensing situations when object size and texture vary greatly. Especially suited for practical implementation in large-scale or long-term monitoring systems, ASPP SegNet can better balance accuracy and generalization utilizing contextual awareness.

Visual and quantitative evaluations confirm that ASPP SegNet is this work's most dependable and scalable design for glacial lake segmentation. Especially in data-rich but computationally limited situations, its resilience against spectral ambiguity, fractured lake forms, and terrain interference shows its promise as a basic component of glacier monitoring pipelines. Furthermore, numerous directions show themselves for the next investigation. Initially, temporal Sentinel-2 time series might enable early identification of possible flood hazards and dynamic monitoring of lake dynamics and coastal areas. Second, by combining optical and Synthetic Aperture Radar (SAR) data, constraints related to cloud cover, snow, or low-illumination situations might be addressed, hence strengthening segmentation robustness. Third, expanding the present binary framework to a multi-class configuration might help to map various surface types (e.g., moraine, trash, snow, water) for more complete glacier and coastal landscape characterization.

Overall, this work adds to the mounting corpus of work at the junction of environmental remote sensing and deep learning. It shows the viability and need of choosing suitable model architectures for geospatial applications. It offers a useful benchmark for further work on climate resilience, water resource management, and catastrophe risk reduction in mountainous and cryospheric areas. While this study focused on static image-based segmentation, glacial lakes are inherently dynamic, with seasonal and long-term morphological changes driven by climatic variability. Future work should therefore evaluate the temporal robustness of segmentation models by applying them to multi-date satellite imagery across different seasons and years. Such testing would reveal the ability of models to generalize under varying illumination, snow cover, turbidity, and melt conditions, which are critical for reliable long-term monitoring. Incorporating temporal modeling strategies, such as recurrent neural networks or time-series data augmentation, could further enhance the adaptability of deep learning frameworks for dynamic cryospheric applications.

Furthermore, considering the trade-off between computational efficiency and segmentation accuracy is critical for operational deployments. While ASPP SegNet demonstrates the highest resilience and generalization, its multiscale architecture demands greater computational resources, making it more suitable for offline or cloud-based processing. In contrast, the Simple CNN, owing to its lightweight design and faster inference times, could be advantageous for real-time glacial lake monitoring applications where computational resources are limited, such as on-board satellite systems or field-based mobile platforms. U-Net offers an intermediate solution, balancing accuracy and computational demand. Future implementations could explore model compression, pruning, or lightweight architectural variants to optimize segmentation performance under practical constraints, thereby enhancing the applicability of deep learning models in glacial hazard early-warning systems.

V. Acknowledgement

The authors extend their appreciation to Northern Border University, Saudi Arabia, for supporting this work through project number (NBU-CRP-2025-2461).

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This article has been accepted for publication in IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing. This is the author's version which has not been fully e
content may change prior to final publication. Citation information: DOI 10.1109/JSTARS.2025.3576187

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This article has been accepted for publication in IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing. This is the author's version which has not been fully e
content may change prior to final publication. Citation information: DOI 10.1109/JSTARS.2025.3576187

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